

Cyclotomy

Cyclotomy is the study of the “roots of unity”.

Given a positive integer n , an n -th root of unity in a commutative ring R with identity is an element ζ such that $\zeta^n = 1$. In other words, ζ is a root in R of the polynomial $X^n - 1$.

If n is the *smallest* positive integer such that $\zeta^n = 1$, then we say that ζ is a *primitive* n -th root of unit in R . We will be interested in defining and studying a polynomial $\Phi_n(X)$ such that primitive n -roots of unity are roots of this polynomial.

Some properties of $X^n - 1$

If d divides n , then we note that

$$\frac{X^n - 1}{X^d - 1} = \frac{(X^d)^{n/d} - 1}{(X^d) - 1} = \sum_{k=0}^{\frac{n}{d}-1} (X^d)^k = (X^d)^{\frac{n}{d}-1} + \cdots + (X^d) + 1$$

This shows that $X^d - 1$ divides $X^n - 1$.

It follows that if $d = \gcd(m, n)$, then $X^d - 1$ divides $\gcd(X^m - 1, X^n - 1)$.

We note that for all k and n , we have $\gcd(X^k, X^n - 1) = 1$.

Now, if $d = \gcd(m, n)$, there are positive integers p and q such that $d = mp - nq$. It follows that $X^{mp} - X^{nq} = X^{nq}(X^d - 1)$. If $P = \gcd(X^m - 1, X^n - 1)$ in $\mathbb{Q}[X]$, then P divides $X^m - 1$ and thus also $X^{mp} - 1$. Similarly P divides $X^{nq} - 1$. It follows that P divides $X^{mp} - X^{nq}$. Now, $\gcd(P, X^{nq})$ divides $\gcd(X^{nq}, X^n - 1) = 1$; this shows that $\gcd(P, X^{nq}) = 1$. It follows that P divides $X^d - 1$. Hence, we have proved:

If $d = \gcd(m, n)$, the $X^d - 1 = \gcd(X^m - 1, X^n - 1)$ in $\mathbb{Q}[X]$.

The formal derivative $D(X^n - 1)$ is nX^{n-1} . It follows that if n is invertible in a field F , then $\gcd(X^n - 1, D(X^n - 1)) = 1$ in $F[X]$. As seen earlier, this means that $X^n - 1$ has no square factors. In other words, if $X^n - 1 = P(X)Q(X)$, then $\gcd(P(X), Q(X)) = 1$ in $F[X]$.

In particular, this is true for $F = \mathbb{Q}$. This means that

$$X^n - 1 = \prod_{P(X) \in S_n} P(X)$$

where S_n consists of *distinct* irreducible polynomials in $\mathbb{Q}[X]$. Moreover, by what has been proved earlier, we can assume that each $P(X)$ is a *monic* polynomial in $\mathbb{Z}[X]$ and in that case each $P(X)$ is uniquely determined, so $S_n \subset \mathbb{Z}[X]$ is well-defined. Since $X^d - 1$ divides $X^n - 1$ when d divides n , we see that $S_d \subset S_n$. Moreover, if $d = \gcd(m, n)$, we have $S_d = S_m \cap S_n$.

The Cyclotomic polynomial $\Phi_n(X)$

With notation as above, choose a $P(X)$ in S_n . Then $\mathbb{Q}[X]/\langle P(X) \rangle$ is a (finite) field extension of $\mathbb{Q}$ and $Z_P = \mathbb{Z}[X]/\langle P(X) \rangle$ is a domain inside it. Let x_P be the image of X in Z_P , so that $P(x_P) = 0$. Since $P(X)$ divides $X^n - 1$, we see that $x_P^n = 1$ in the domain Z_P . Let $d(P)$ denote the multiplicative order of x_P . Note that $1, x_P, \dots, x_P^{d(P)-1}$ are distinct elements of Z_P , and $d(P)$ divides n .

We see that S_n is the disjoint union of the sets $T_d = \{P(X) : d(P) = d\}$ where d divides n . Since $S_d \subset S_n$ in this case, we see that the definition of T_d does not depend on n . We define the n -th cyclotomic polynomial

$$\Phi_n(X) = \prod_{P(X) \in T_n} P(X)$$

Note that

$$T_n = S_n \setminus \bigcup_{d|n; d < n} T_d = S_n \setminus \bigcup_{d|n; d < n} S_d$$

so that $\Phi_n(X)$ can be directly defined (without factorisation) as the greatest common divisor of the polynomials $(X^n - 1)/(X^d - 1)$ where d divides n and is less than n :

$$\Phi_n(X) = \gcd \left\{ \frac{X^n - 1}{X^d - 1} \mid d < n; d|n \right\}$$

in $\mathbb{Q}[X]$. As usual, when we require that $\Phi_n(X)$ is a monic polynomial in $\mathbb{Z}[X]$ it is uniquely determined by this equation. We also note from the above that

$$X^n - 1 = \prod_{d|n} \Phi_d(X)$$

and note that this means that n is the sum of the degrees of Φ_d as d varies over positive integers dividing n .

Suppose D is a domain which contains ζ which is a primitive n -th root of unity. (For example, we can take $D = Z_P$ for $P \in T_n$ and $\zeta = x_P$). Then $\zeta, \dots, \zeta^{n-1}, \zeta^n = 1$ are n distinct roots of $X^n - 1$. Thus, $X^n - 1 = \prod_{k=1}^n (X - \zeta^k)$ in $D[X]$. Further note that if $\gcd(k, n) = e$, then ζ^k has order $n/e = d$. Hence, if $P(\zeta^k) = 0$ and $P \in S_n$, then $P \in T_d$. It follows that

$$\Phi_n(X) = \prod_{\substack{0 < k \leq n \\ \gcd(k, n) = 1}} (X - \zeta^k)$$

We define the Euler totient $\varphi(n)$ as the cardinality of the finite set $\{0 < k \leq n : \gcd(k, n) = 1\}$. Then the degree of $\Phi_n(X)$ is $\varphi(n)$.

Irreducibility of $\Phi_n(X)$

We will now show that $\Phi_n(X)$ is irreducible. In other words, $T_n = \{\Phi_n(X)\}$ and $S_n = \{\Phi_d(X) : d \text{ divides } n\}$, so that

$$X^n - 1 = \prod_{d|n} \Phi_d(X)$$

is the decomposition of $X^n - 1$ into irreducible factors in $\mathbb{Z}[X]$.

Given positive integer k and n we have seen that $S_n \subset S_{kn}$. This means that for every element $P(X)$ in S_n , there is an element $Q(X)$ in S_n such that $P(X)$ divides $Q(X^k)$ (which is a factor of $X^{kn} - 1$).

Now, suppose in addition that $\gcd(k, n) = 1$ and d divides n . In this case, we have $S_d = S_n \cap S_{kd}$. This means that if $Q(X)$ is in S_d , so that $Q(X^k)$ is a factor of $X^{kd} - 1$, then $P(X)$ must be in S_d . Since $S_d \cap T_n$ is empty if $d < n$, it follows that if $P(X) \in T_n$ then $Q(X)$ cannot be in S_d for $d < n$. In other words, if $P(X)$ is in T_n , then $Q(X)$ is also in T_n .

For every k co-prime to n , and $P(X) \in T_n$, there is a $Q(X)$ in T_n such that $P(X)$ divides $Q(X^k)$.

Choose a prime p which does not divide n . Since all the polynomials in S_n have coefficients in $\mathbb{Z}$, we can reduce modulo p . Note that for any polynomial $R(X)$ with coefficients in $\mathbb{Z}/\langle p \rangle$ we have $R(X^p) = R(X)^p$.

Since p is co-prime to n , for very $P(X)$ in T_n , there is a $Q(X)$ in T_n such that $P(X)$ divides $Q(X^p)$. It follows that $P(X) \pmod{p}$ divides $Q(X)^p \pmod{p}$.

Since n is invertible modulo p , we see that $X^n - 1$ has distinct irreducible factors modulo p . Thus, $P(X)$ and $Q(X)^p$ can have a common factor modulo p if and only if $P(X) = Q(X)$.

For every prime p that does not divide n and $P(X) \in T_n$, we have $P(X)$ divides $P(X^p)$.

Note that this means that $P(X^k)$ divides $P(X^{kp})$ for every $P(X) \in T_n$ and for every positive integer k .

If $\gcd(k, n) = 1$, then k can be written as a product of primes p which do not divide n . Thus, by induction on the number of such prime factors, we see that $P(X)$ divides $P(X^k)$ for every $P(X)$ in T_n .

It follows that $P(X) \in T_n$ has the roots $\{x_P^k \mid 0 < k \leq n; \gcd(k, n) = 1\}$ in the domain Z_P . It follows that the degree of $P(X)$ is $\varphi(n)$. Hence $P(X) = \Phi_n(X)$ and $T_n = \{\Phi_n(X)\}$.

Some calculations

A 1-st root of unity is 1! It is a root of $\Phi_1(X) = X - 1$.

When $n = p$ is a prime, we see that

$$\Phi_p(X) = \frac{X^p - 1}{X - 1} = X^{p-1} + \cdots + X + 1$$

Similarly, if $n = p^r$, then one sees that

$$\Phi_{p^r}(X) = \frac{X^{p^r} - 1}{X^{p^{r-1}} - 1} = \Phi_p(X^{p^{r-1}})$$

One shows, by induction on n that if $n > 1$, then $\Phi_n(0) = 1$ for $n \geq 2$.

Another useful observation is that the sum of *all* primitive n -th roots of unity is a coefficient in $\Phi_n(X)$; hence, it is an integer.

Primes congruent to 1 modulo n

By the above, if N is *any* integer, then $\Phi_n(N)$ is congruent to 1 modulo N . Thus, $\Phi_n(N)$ is co-prime with N .

Suppose p is a prime which does not divide n . This implies that $X^n - 1$ has distinct factors in $(\mathbb{Z}/\langle p \rangle)[X]$. In particular, $\Phi_n(X)$ does not have a common root with $\Phi_d(X)$ for any d dividing n .

Suppose p also divides $\Phi_n(N)$. Then $N \pmod{p}$ gives a primitive n -th root of unity modulo p .

Since the non-zero integers modulo p give a group of order $(p - 1)$ and N gives an element of order n in this group, we see that n divides $p - 1$. Equivalently, p is congruent to 1 modulo n . In particular, we see that ζ_n can be seen as an element of $\mathbb{Q}_p$. In fact, in terms of the function $\omega : (\mathbb{Z}/\langle p \rangle)^* \rightarrow \mathbb{Z}_p$ defined earlier, we can see that $\zeta_n = \omega(N) = \lim_{k \rightarrow \infty} N^{p^k}$ in $\mathbb{Z}_p$.

Given any finite collection $\{p_1, \dots, p_r\}$ of primes, take $N = n \cdot p_1 \cdots p_r$. Then $\Phi_n(N)$ is co-prime to all the primes p_i and co-prime to n . Hence, a prime factor p of $\Phi_n(N)$ is *distinct* from all the p_i and is co-prime to n .

Given any finite collection of primes, there is a prime p distinct from all of these such that p is congruent to 1 modulo n .

This shows that there are infinitely primes of this type!

The ring $\mathbb{Z}[\zeta_n]$

By the above, we see that there *is* a primitive n -root of unity ζ_n in $\mathbb{Q}_p$ for suitable choices of p . Thus, the ring of cyclotomic integers can be considered as the smallest subring of $\mathbb{Q}_p$ that contains ζ_n and 1.

Alternatively, we see that $\zeta_n^a = \exp\left(\frac{2a\pi\sqrt{-1}}{n}\right)$ are n -roots of unity in the field $\mathbb{C}$ of complex numbers. Thus, the ring of cyclotomic integers can be considered as the smallest subring of $\mathbb{C}$ that contains ζ_n and 1.

We will now fix n and one such choice of ring $\mathbb{Z}[\zeta_n]$. To simplify notation we will use ζ to denote a primitive n -th root of unity.

Note that the elements ζ^b as b varies over integers form a cyclic group of order n under multiplication. It follows that ζ^b has order $d = n / \gcd(b, n)$ by standard results in group theory. Thus, it is primitive d -th root of unity. Moreover, if a is such that $\gcd(a, n) = 1$, then ζ^{ab} is also a primitive d -root of unity since $\gcd(ab, n) = \gcd(b, n)$ in that case.

In particular, we see that $\mathbb{Z}[\zeta_n]$ contains $\mathbb{Z}[\zeta_d]$ for d dividing n .

Units in $\mathbb{Z}[\zeta_n]$

It is obvious that ζ^k are units in the ring $\mathbb{Z}[\zeta_n]$.

When a is a positive integer, we note that

$$\frac{1 - \zeta^a}{1 - \zeta} = 1 + \zeta + \cdots + \zeta^{a-1}$$

is an element of $\mathbb{Z}[\zeta_n]$.

If a is an integer such that $\gcd(a, n) = 1$, then there are integers b and c such that $ab + nc = 1$; moreover, we can assume that b is positive. It follows that

$$(\zeta^a)^b = \zeta^{1-nc} = \zeta$$

It follows that

$$\frac{1 - \zeta}{1 - \zeta^a} = \frac{1 - (\zeta^a)^b}{1 - (\zeta^a)} = 1 + (\zeta^a) + \cdots + (\zeta^a)^{b-1}$$

is also an element of $\mathbb{Z}[\zeta_n]$.

When $\gcd(a, n) = 1$, we have a unit $(1 - \zeta^a)/(1 - \zeta)$ in $\mathbb{Z}[\zeta_n]$.

This group generated by such units is known as the *group of cyclotomic units*.

Automorphisms of $\mathbb{Z}[\zeta_n]$

If $\gcd(a, n) = 1$, then we have seen above that there is a positive integer b such that $(\zeta^a)^b = \zeta$.

It follows easily that if $\zeta^{am} = 1$, then $\zeta^m = 1$. Thus, if ζ is a primitive n -th root of unity then so is ζ^a .

In other words, the algebraic properties of ζ and ζ^a are identical. This can be used to show that that

$$\sum_{k=0}^{n-1} c_k \zeta^k \mapsto \sum_{k=0}^{n-1} c_k \zeta^{ak}$$

is a ring automorphism of $\mathbb{Z}[\zeta_n]$.

If an element $\sum_{k=0}^{n-1} c_k \zeta^k$ is *invariant* under the above automorphisms, then $c_{ak} = c_k$ for each a such that $\gcd(a, n) = 1$. For each d dividing n , we note

$$\begin{aligned} \{k \in \mathbb{Z}/\langle n \rangle : \gcd(k, n) = d\} \\ &= \{ad \in (\mathbb{Z}/\langle n \rangle) : \gcd(a, n) = 1\} \\ &= \{ad \in (\mathbb{Z}/\langle n \rangle) : \gcd(a, (n/d)) = 1; 1 \leq a < n/d\} \end{aligned}$$

Thus see that $c_d = c_{ad}$ for all a lying between 1 and $(n/d)-1$ with $\gcd(a, n/d) = 1$. Reversing the roles of d and n/d for notational convenience, we have

$$\sum_{k=0}^{n-1} c_k \zeta^k = \sum_{d|n} c_{n/d} \sum_{\substack{a=1 \\ \gcd(a, d)=1}}^{d-1} \zeta^{a(n/d)}$$

Moreover, $\zeta^{a(n/d)}$ is a primitive d -th root of unity for all a such that $\gcd(a, d) = 1$, and as a varies, we get *all* such primitive roots. As seen above, this means that $\sum_{\gcd(a, d)=1; a=1}^{d-1} \zeta^{a(n/d)}$ is an integer.

The elements of $\mathbb{Z}[\zeta_n]$ that are *fixed* under *all* these automorphisms is precisely the subring $\mathbb{Z}$ of integers.

We also note that we have the identity

$$\Phi_n(X) = \prod_{a \in (\mathbb{Z}/\langle n \rangle)^*} (X - \zeta^a)$$

in the ring $\mathbb{Z}[\zeta_n]$ since the elements ζ^a for $\gcd(a, n) = 1$ give precisely the collection of roots of $\Phi_n(Z)$.

Divisors of p

Let us now specialise to the case where $n = p$ is an odd prime number. In that case, we have

$$\Phi_p(X) = X^{p-1} + \cdots + X + 1 = \prod_{a=1}^{p-1} (X - \zeta^a)$$

Substituting $X \mapsto 1$, we get the identity

$$p = \prod_{a=1}^{p-1} (1 - \zeta^a) = (1 - \zeta) \cdot (1 - \zeta^2) \cdots (1 - \zeta^{p-1})$$

We have seen that $(1 - \zeta^a)/(1 - \zeta) = u_a$ is a unit in $\mathbb{Z}[\zeta_p]$ for these values of a . It follows that

$$p = (1 - \zeta)^{p-1} u_2 \cdots u_{p-1}$$

Since $p-1$ is divisible by 2, we could wonder whether there is a square root of $\pm p$ hidden in the ring $\mathbb{Z}[\zeta_p]$.

Square roots and automorphisms

For a quadratic equation $X^2 + aX + b$, if α is a root, then so is $-a - \alpha$. Specifically, when $a = 0$, we see that if α is a root then so is $-\alpha$.

Thus, one way to attempt to identify a square root in $\mathbb{Z}[\zeta_p]$ would be to find an element α that is mapped to $\pm\alpha$ under the automorphisms of this ring.

$p = 3$. In this case, we note that the automorphism is $\zeta \mapsto \zeta^{-1} = \zeta^2$. Thus, it is natural to consider the element $\alpha = \zeta - \zeta^2$. Using the formula $\Phi_3(X) = X^2 + X + 1$ we obtain

$$\alpha^2 = \zeta^2 + \zeta - 2 = (\zeta^2 + \zeta + 1) - 3 = -3$$

$p = 5$. In this case, we note that the automorphisms form a cyclic group of order 4 generated by $\zeta \mapsto \zeta^2$. So, if this takes α to $-\alpha$, then $\zeta \mapsto \zeta^4$ takes α to itself. Thus, we take $\alpha = \zeta - \zeta^2 - \zeta^3 + \zeta^4$. We use the formula $\Phi_5(X) = X^4 + X^3 + X^2 + X + 1$ to calculate

$$\alpha^2 = \zeta^2 + \zeta^4 + \zeta + \zeta^3 + 2(-\zeta^3 - \zeta^4 + 1 + 1 - \zeta - \zeta^2) = -1 + 2 \cdot 3 = 5$$

$p = 7$. We note that the automorphisms form a cyclic group of order 6 generated by $\zeta \mapsto \zeta^3$. Mimicking the idea above, we write

$$\alpha = \zeta - \zeta^3 + \zeta^2 - \zeta^6 + \zeta^4 - \zeta^5$$

One can then check that $\alpha^2 = -7$.

Gauss sum

The basic idea is as follows. For each a such that $\gcd(a, p) = 1$, we guess that the sign attached by the automorphism $\zeta \mapsto \zeta^a$ is a *homomorphism* $(\mathbb{Z}/\langle p \rangle)^* \rightarrow \{\pm 1\}$. Since the former is a cyclic group of order $(p-1)$, this homomorphism is given by $a \mapsto a^{(p-1)/2}$. With this intuition, Gauss defined

$$G(p) = \sum_{a=1}^{p-1} (a^{\frac{p-1}{2}} \bmod p) \zeta^a$$

as an element of $\mathbb{Z}[\zeta_p]$. The claim is that $G(p)^2 = (-1)^{\frac{p-1}{2}} p$.

To prove the claim we replace ζ by a variable T (which satisfies $T^p = T$) to get

$$g(T) = \sum_{a=1}^{p-1} (a^{\frac{p-1}{2}} \bmod p) T^a$$

We then calculate

$$g(T)^2 = \sum_{a=1}^{p-1} \sum_{b=1}^{p-1} ((ab)^{\frac{p-1}{2}} \bmod p) T^{a+b} = \sum_{k=0}^{p-1} c_k T^k$$

for some integers c_k , where we have replaced T^{a+b} by $T^{a+b \bmod p}$.

Now, look at the terms which contribute for T^0 , we get $c_0 = \sum_{a=1}^{p-1} ((-a^2)^{\frac{p-1}{2}} \bmod p)$. Since $(a^2)^{\frac{p-1}{2}} = 1$, we see that $c_0 = (-1)^{(p-1)/2}(p-1)$.

Next, look at the terms which contribute to c_k (i.e. pairs (a, b) with $a+b = k \neq 0$). Dividing by k in $\mathbb{Z}/\langle p \rangle$, we see that the pair $((a/k), (b/k))$ contributes to the term c_1 . We further note that, since $(k^2)^{\frac{p-1}{2}} \bmod p$ is 1, we have

$$\left(\frac{a}{k} \cdot \frac{b}{k}\right)^{\frac{p-1}{2}} \bmod p = (ab)^{\frac{p-1}{2}} \bmod p$$

Thus, $c_k = c_1 = \sum_{a=2}^{p-1} ((a(1-a))^{\frac{p-1}{2}} \bmod p)$ for all $k \neq 0$ in $\mathbb{Z}/\langle p \rangle$.

In other words, we have

$$g(T)^2 = (-1)^{\frac{p-1}{2}}(p-1) + c_1 \sum_{k=1}^{p-1} T^k$$

When $T = 1$, we see that $g(T) = 0$ since there are an equal number of positive and negative signs. This gives us $0 = (-1)^{\frac{p-1}{2}}(p-1) + c_1(p-1)$. Thus, $c_1 = -(-1)^{\frac{p-1}{2}}$.

On the other hand, substituting $T = \zeta$, we obtain

$$G(p)^2 = (-1)^{\frac{p-1}{2}}(p-1) + c_1 \sum_{k=1}^{p-1} \zeta^k = (-1)^{\frac{p-1}{2}} \left(p-1 - \sum_{k=1}^{p-1} \zeta^k \right)$$

Using the formula $\Phi_p(X) = X^{p-1} + \cdots + X + 1$, we see that $\sum_{k=1}^{p-1} \zeta^k = -1$. This gives us the required identity.